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BACHELOR OF SCIENCE IN APPLIED DATA SCIENCE COMMUNICATION

ASSIGNMENT - 03

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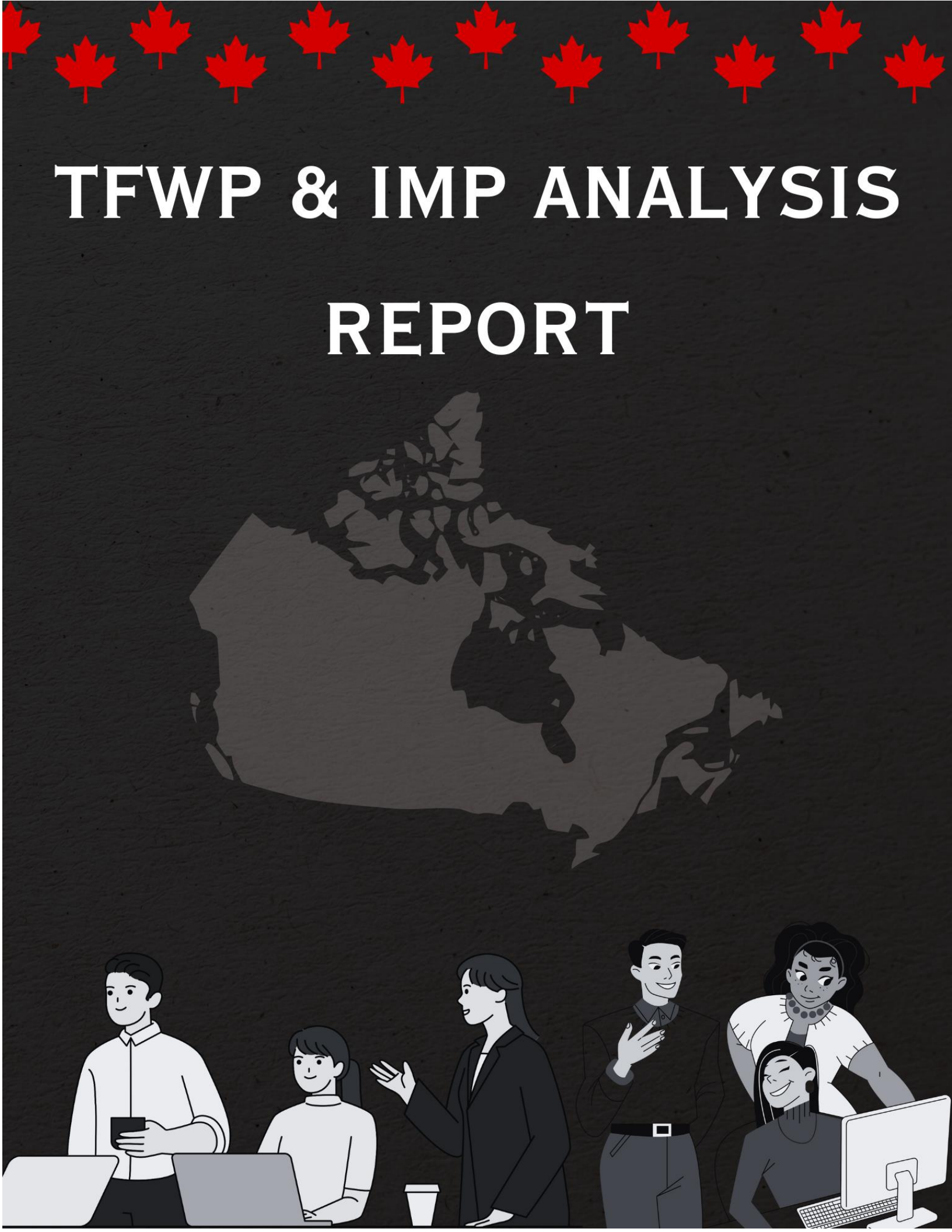
TFWP & IMP ANALYSIS REPORT



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1. Introduction

The Temporary Foreign Worker Program (TFWP) and International Mobility Program (IMP) play crucial roles in addressing labor market demands in Canada. Through various facilitations of the employment of foreign workers. These programs give essential information on regional labor needs and occupational trends, together with socio-economic patterns, in order to understand how the dynamic of the workforce varies. This report focuses on analyzing data from the Canadian Monthly IRCC Updates dataset to explore trends and patterns in worker inflows under these programs.

2. Preparation of data

2.1 Data Source & Extraction

The dataset used to visualize our dashboard was obtained from the Canadian Monthly IRCC Updates, issued by Immigration, Refugees, and Citizenship Canada (IRCC) website : https://open.canada.ca/data/en/dataset/360024f2-17e9-4558-bfc1-3616485d65b9 that include public datasets containing information on temporary foreign worker program and international mobility program. We downloaded the dataset which consists data that was spread across 10 years from 2015 to 2024. They contain monthly data, quarterly data and yearly data of every occupation in this program categorized for every province / territory of Canada.

(Write the excel sheet cleaning part) Then we imported the dataset into SQL Server Management System using the import flat file option. We had to change the data type of all columns to int and it was a challenge since there were more than 50 columns.

2.2 Data Cleaning

Then we had to clean the data in our table so that there are no null values and duplicates.

We used the following code to remove duplicates;

```
WITH CTE_Duplicates AS (
   SELECT *,
         ROW_NUMBER() OVER (
            PARTITION BY Province/Territory and Occupation, Year, Month …….
   FROM dbo.TFWPandIMP
```

```
)
DELETE FROM CTE_Duplicates
WHERE RowNum > 1;
```

- CTE-common table expression; it is a temporary result set, which is named and derived from a simple query. It's defined within the scope of a single SELECT, INSERT, UPDATE, or DELETE statement. This query removes duplicate rows by keeping only the first occurrence of each unique row. The ROW_NUMBER() function assigns a unique row number to each duplicate row in partitions based on the specified columns. Rows with a row number greater than 1 are considered duplicates and are deleted, leaving only one unique row for each set of duplicate data.

When removing null values, we set the null values to ‘N/A’ for the columns with the data type Nvarchar and ‘0’ to the columns with the data type – Int.

The simple code is as below;

```
UPDATE dbo.IFWPandIMP
SET Province/Terriotory_and_Occupation = 'N/A'
WHERE Province/Terriotory_and_Occupation IS NULL;
```

After cleaning the data, we backed up our database to a bak file. Then we opened Power BI and imported the table by connecting our SQL database to Power BI.

This dataset was then transformed in Power BI itself, creating several conditional and custom columns. Using the Split Column by Delimiter

feature, the Province/Territory & Occupation column, which contained combined information, was separated into Province/Territory Code and Occupation. Since some rows contained totals or "N/A," the Province/Territory Code column was checked for the presence of the word "Total" and a Conditional Column was added to categorize these rows in order to tell totals apart from Detail rows. To fill in the empty rows in the Occupation column, Fill Down transformation was applied to get the last valid value into empty cells.

The Attribute column consisted of time-related data, for instance, 11_2015, Q1_Total_2015, 2015Total, which was parsed into Year, Month, and Quarter columns via a tailor-made logic. For example, the rows containing “Q” were grouped as quarterly data, and the rows ended with “Total” were classified as yearly data. Moreover, a final classification column was added to separate Monthly Data, Quarterly Data and Yearly Data. The transformations made sure the dataset was clean, structured correctly, and allowed for accurate visualizations and insights.

3. Visualisation findings

3.1 User Interface & Functionality

The dashboard has a user-friendly interface that easy for navigation. The first page which is the home page has a dashboard design where the user can choose which page or section they want to see. By clicking on an option, it takes the user to their chosen page. On the top of the page there is a filter icon that you can click on where you get a pop-up filter pane where you could use filters to choose the year, quarter, month, province and the occupation for easier access to data in a specific region or a specific year. You can reset everything to the original state by clicking on the reset icon in the filter pane. You can navigate through the pages by clicking on the page navigation buttons on top of each page.

We used DAX for making almost all the cards in dashboard. We used it to take percentages of the most common outcomes. We have applied filters for almost all the visualizations to get clean and meaningful insights. It is through the combination of these navigation options, filtering capabilities and detailed view functionality, that the user will be able to cover and analyze the data presented on the dashboard effectively and efficiently.

3.2 DAX codes used for the cards:

```
Total_Occupations =
CALCULATE(
   SUM('TFWPandIRCC'[Total]),
   NOT(CONTAINSSTRING('TFWPandIRCC'[Province/Territory and Occupation], "Total"))
)
```

- *To get the total number of occupations throughout all the provinces*

```
Total_Current_Year =
CALCULATE(
   SUMX(
      FILTER(
         'TFWPandIRCC',
         'TFWPandIRCC'[Year] = MAX('TFWPandIRCC'[Year]) &&
         NOT(CONTAINSSTRING('TFWPandIRCC'[Province/Territory and Occupation], "Total"))
      ),
      'TFWPandIRCC'[Total]
   ))
```

- *To get the total number of occupations in the current year*

```
Overall_Growth_Percentage =
VAR StartYear = MINX(
```

```
    FILTER('TFWPandIRCC',
ISNUMBER(VALUE('TFWPandIRCC'[Year]))),
    VALUE('TFWPandIRCC'[Year])
)
VAR EndYear = MAXX(
    FILTER('TFWPandIRCC',
ISNUMBER(VALUE('TFWPandIRCC'[Year]))),
    VALUE('TFWPandIRCC'[Year])
)
VAR StartYearTotal =
    CALCULATE(
        SUM('TFWPandIRCC'[Total]),
        'TFWPandIRCC'[Year] = FORMAT(StartYear, "0"),
        NOT(CONTAINSSTRING('TFWPandIRCC'[Province/Territory and
Occupation], "Total"))
    )
VAR EndYearTotal =
    CALCULATE(
        SUM('TFWPandIRCC'[Total]),
        'TFWPandIRCC'[Year] = FORMAT(EndYear, "0"),
        NOT(CONTAINSSTRING('TFWPandIRCC'[Province/Territory and
Occupation], "Total"))
    )
RETURN
```

```
IF(
    StartYearTotal = 0,
    BLANK(),
    DIVIDE(EndYearTotal - StartYearTotal, StartYearTotal, 0) * 100)
```

- This measure is used to get the overall growth through the years. VALUE function is used for a number, or a formula that evaluates to a numeric value. FORMAT is used for a number format that you specify.

3.3 Visualisation findings

3.3.1 Count Distinct of Total of Top 20 Occupations

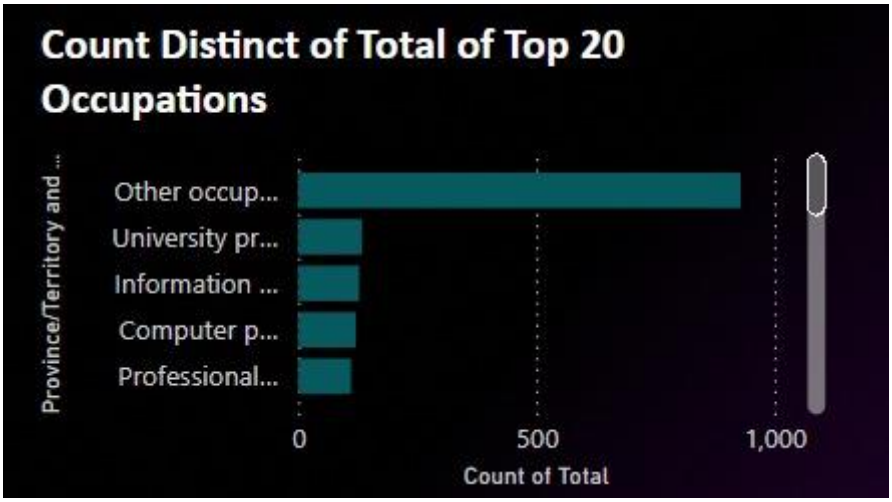

Figure 2

- The "Count Distinct of Total of Top 20 Occupations" bar chart shows the distribution of occupations in Canada, with "Other Occupations" being significantly higher, followed by specific categories like "University Professors" and "Information Professionals."
- Insights indicate that niche categories have a smaller share compared to broader occupational groups.

3.3.2 Count of Distinct Top 15 Jobs by Year and Occupation

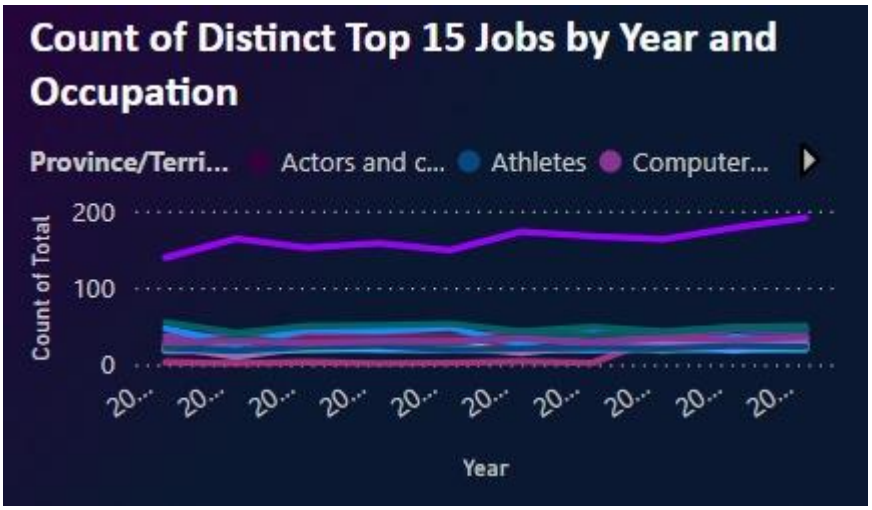

Figure 3

- Show different occupations exhibit varied trends over the years.
- Actors and athletes show moderate increases.
- Computer programmers remain relatively steady.
- This chart provides a detailed view of how certain occupations evolve over time.
- Actors and Comedians, Athletes, and Computer and Information Systems Managers being the top occupations

3.3.3 Sum of Jobs by Occupation

Figure 4

- Shows the total number of occupations.
- University professors have the highest total jobs, followed by information professionals and other occupations. This shows the significant presence of highly skilled professionals in the workforce.

3.3.4 KPI cards

Figure 5

1st - This card shows the total count of occupations across all years and provinces/territories

2nd – Total occupations in 2024

3rd - Percentage growth in total occupations from 2015 to 2024, reflecting a substantial increase in demand or participation in the analyzed occupations

3.3.5 Sum of Total Jobs by Province

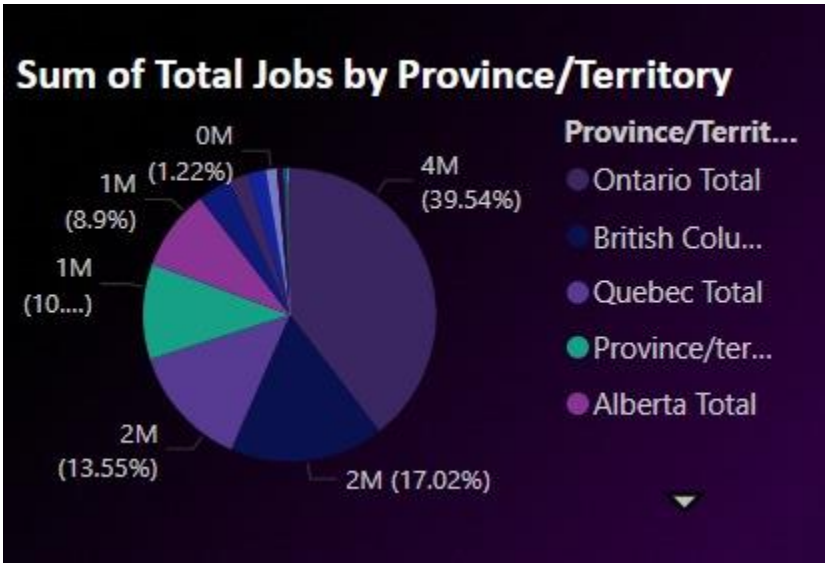

Figure 4

- Ontario accounts for the largest proportion (39.54%) of total jobs.
- Quebec and British Columbia follow with smaller but significant contributions. This shows Ontario's dominance in total employment opportunities.

3.3.6 Count of Distinct Jobs by Year and Province

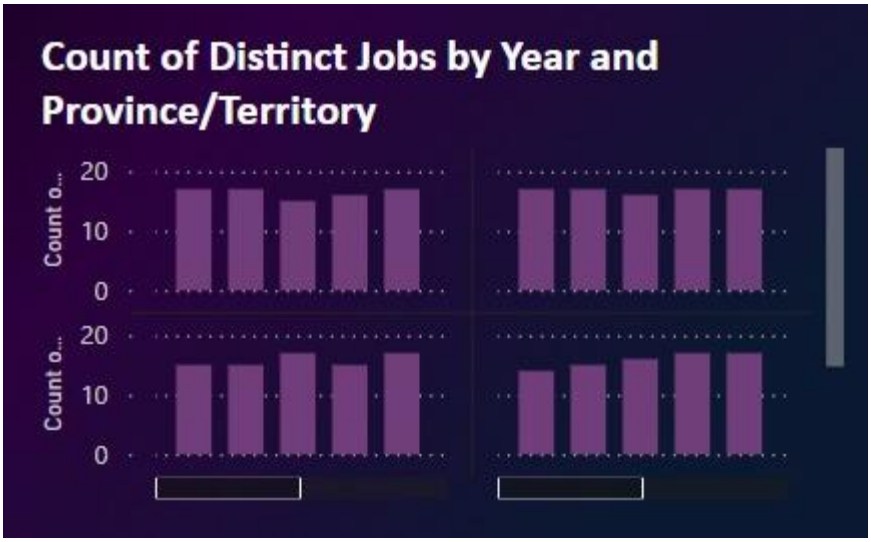

Figure 5

- Different provinces/territories contribute distinct counts of jobs, but the pattern is consistent year over year, with no dramatic fluctuations with Ontario, British Columbia, and Quebec consistently having the highest counts.

3.3.7 Filled Map of Province

Figure 6

3.3.8 Count of Total Distinct Jobs by Year

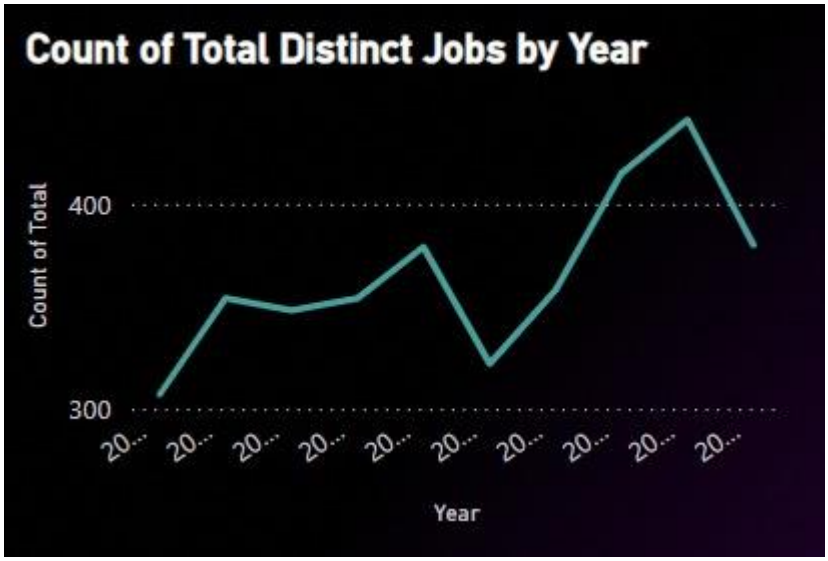

Figure 7

- The count of distinct jobs increases steadily over the years, with some minor dips. This indicates a positive long-term growth trend.

3.3.9 Cumulative Total by Year

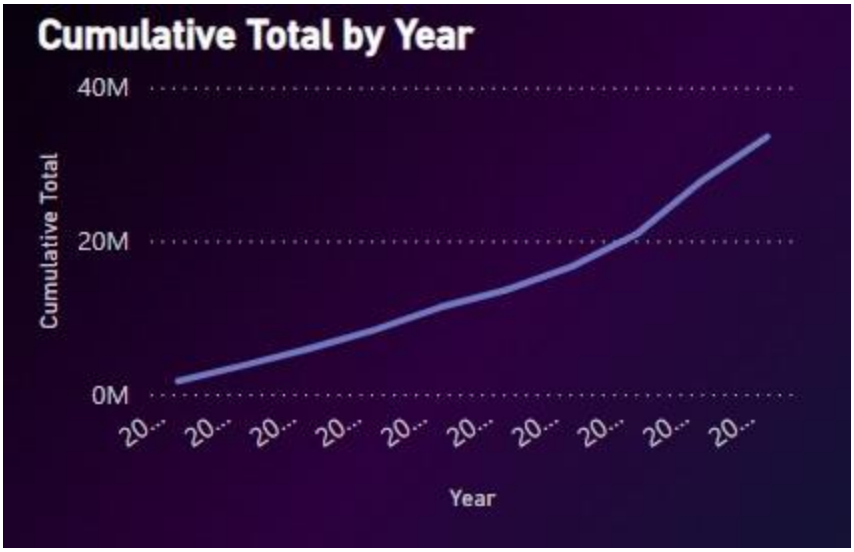

Figure 8

- It reflects a continuous rise in job opportunities as the cumulative total increases through the years.

3.3.10 Sum of total jobs

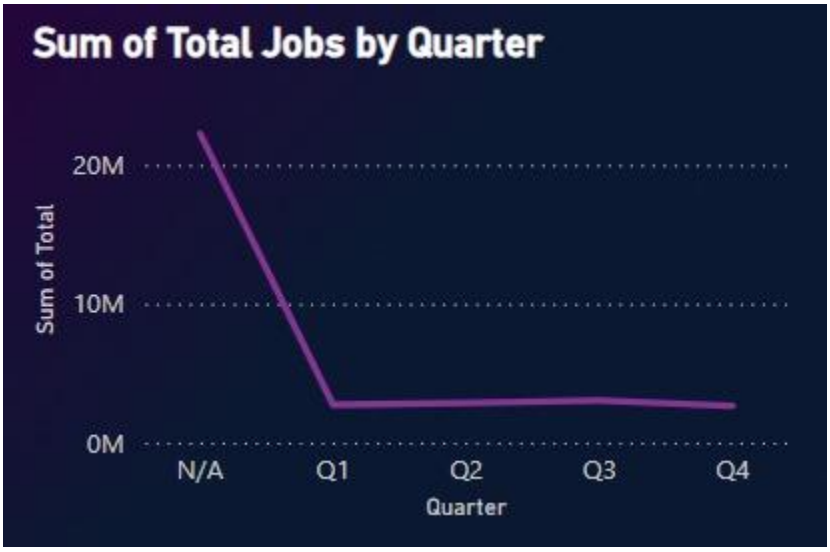

Figure 9

- The distribution shows a steep drop after Q1, indicating either missing data or a sharp seasonal decline in workforce activity after the first quarter.

3.3.11 Sum of Total Jobs by Year

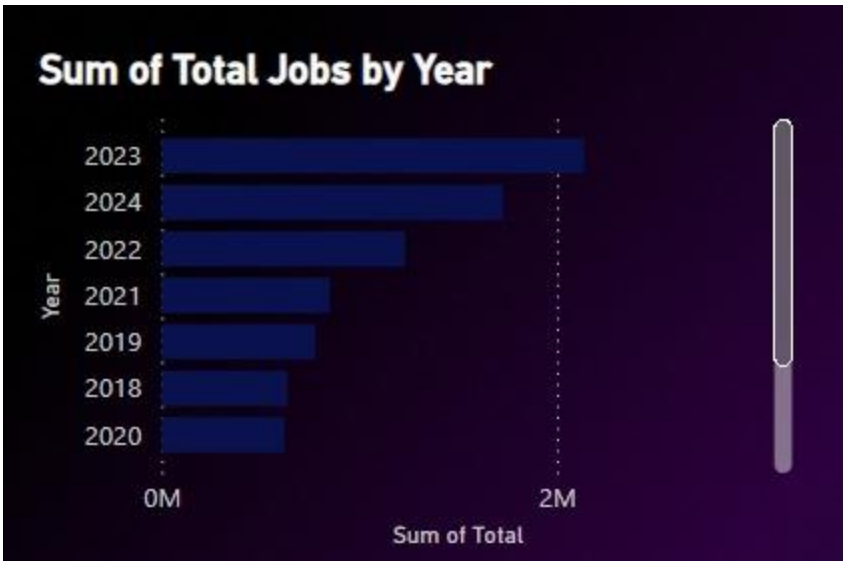

Figure 10

- Total jobs increase gradually from 2015 to 2024, with noticeable peaks in recent years, showing consistent growth in employment

3.3.12 Count of quarter by Total Distinct Jobs and Province/Territory

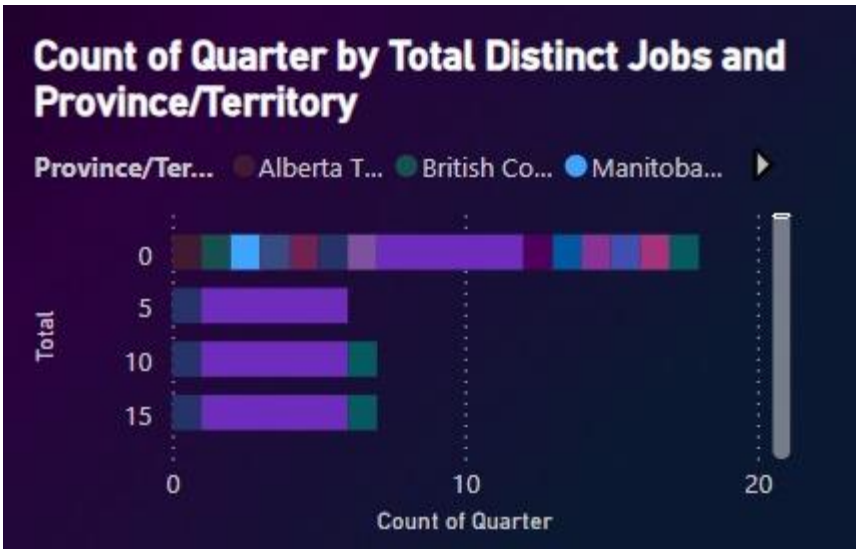

Figure 11

- Alberta, British Columbia, and Manitoba show more uniform distributions that the other provinces but overall there are varying distribution of provinces across all quarters.

3.3.13 Count of Total Distinct Jobs by Month

Figure 12

- The line chart shows seasonal fluctuations in jobs, with notable spikes in certain months like May.

3.3.14 Decomposition Tree

Figure 13

- This graph shows the total number of jobs across Canada over time, from 2016 to 2024.
- The graph indicates a steady increase in total jobs, with a dip in 2020 likely due to the COVID-19 pandemic.
- The largest increase in total jobs is projected to occur between 2023 and 2024.
- This graph shows the total number of jobs for different provinces and territories in Canada over time.
- Ontario consistently has the highest number of jobs, followed by British Columbia and Quebec.
- Alberta, Manitoba, and Saskatchewan also have significant job numbers, but lower than the top three provinces.
- Nova Scotia and New Brunswick have the lowest total jobs among the provinces and territories shown.

3.3.15 Maximum of Total by Occupation and Year

Figure 14

- This graph shows the distribution of jobs across different occupations over time.
- The "Accommodation" occupation has the highest number of jobs, followed by "Accounting" and other occupations.
- The trends for each occupation vary, with some occupations showing an increase in jobs over time, while others remain relatively stable.

3.3.16 Sum of Total Jobs by Year

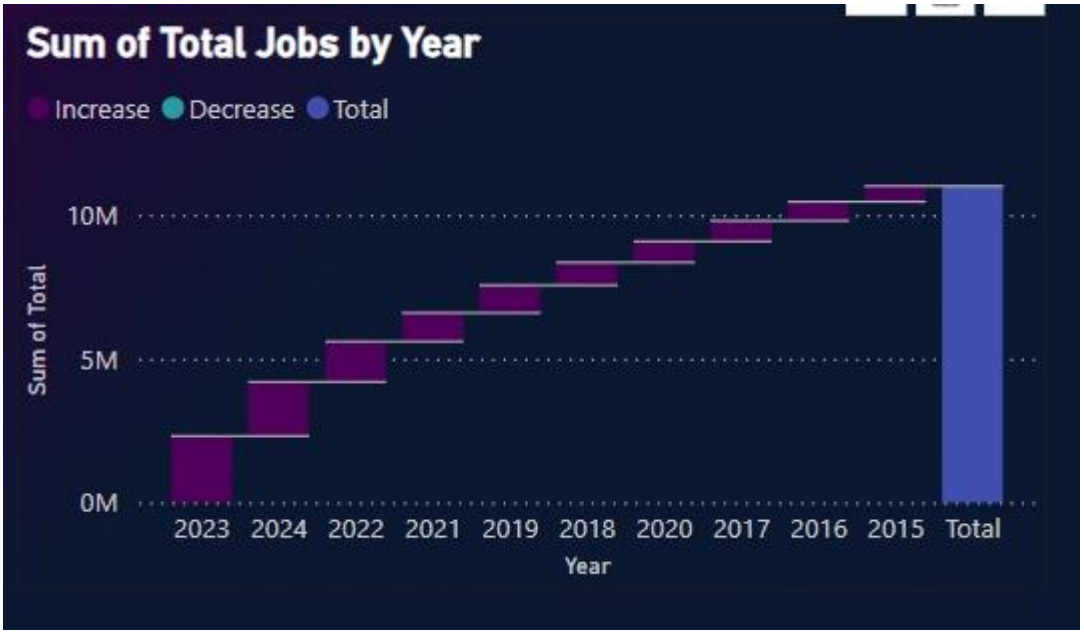

Figure 15

- This graph shows the distribution of jobs across different occupations over time.
- The "Accommodation" occupation has the highest number of jobs, followed by "Accounting" and other occupations.
- The trends for each occupation vary, with some occupations showing an increase in jobs over time, while others remain relatively stable.

4.Conclusion

The analysis of the Temporary Foreign Worker Program (TFWP) and International Mobility Program (IMP) dataset provides valuable insights into occupational trends, regional workforce distribution, and temporal patterns over the past decade. From there, it effectively highlighted key metrics on the dashboard, including the overall growth rate of 246.79% from 2015 to 2024, regional dominance by provinces such as Ontario, and the demand for certain occupations, including university and information professionals. Seasonal trends and quarterly fluctuations further reveal the dynamism of the job market.

The combination of visualizations at a detailed level, including bar charts, pie charts, and line graphs, provides a comprehensive overview of the data. Advanced techniques, including the logarithmic scaling of data and aggregation, make sure that clarity in visual representation is achieved, especially for smaller datasets. These insights could be used to empower the decision-makers in workforce planning, policy adjustments, and areas of improvement. Generally, the report shows the importance of data analysis and visualization in bringing out active intelligence from big data.

Resources

DAX Guide - https://dax.guide/

Tips for IT Pros, SQL Server Remove Duplicate Rows with CTE - https://www.youtube.com/watch?v=6ycNqvaMCzs

Jason Davidson, Creating a Pop-Out Navigation and Filter Bar Using Power BI Bookmarks. 4 Ideas for Sections to Add - https://www.youtube.com/watch?v=SCI9iKl4vjU&t=1096s

SQLBI, Measures vs. calculated columns in DAX and Power BI - https://www.youtube.com/watch?v=ePPi1LLX0sA&list=LL&index=1&t=45s

Kevin Stratvert, How to use Power BI DAX – Tutorial - https://www.youtube.com/watch?v=waG_JhBgUpM&t=341s

Microsoft Learn, https://learn.microsoft.com/en-us/dax/topn-function-dax